

VAL MARIE SOUTHEAST, SK, Canada

WMO: 711370

Lat: 49.060N

Lon: 107.590W

Elev: 2612

StdP: 13.36

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.9	-21.2	-35.2	0.8	-27.8	-27.8	1.3	-21.0	32.0	26.3	29.3	25.6	3.4	60	0.723

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.8	92.3	62.6	88.5	61.7	84.7	60.8	66.8	82.2	64.8	80.5	63.1	78.9	11.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	91.8	72.2	59.6	83.9	69.7	57.5	77.9	67.9	32.9	82.1	31.2	80.7	29.9	78.8	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.5	21.5	-37.4	98.0	6.8	3.3	-42.3	100.3	-46.3	102.3	-50.1	104.1	-55.1	106.5
			-37.5	70.8	6.7	2.9	-42.3	72.9	-46.2	74.6	-50.0	76.2	-54.9	78.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.3	10.5	13.9	26.2	40.7	51.0	59.7	67.1	65.6	55.0	41.1	25.9	12.9
	DBStd	22.87	15.86	15.13	14.36	9.44	8.10	6.46	6.02	6.61	8.19	9.03	12.93	14.63
	HDD50	5574	1226	1011	738	298	86	6	0	1	41	295	723	1149
	HDD65	9633	1691	1431	1203	728	436	182	47	75	313	740	1173	1614
	CDD50	1661	0	0	1	19	119	297	529	483	191	21	1	0
	CDD65	242	0	0	0	0	3	23	111	92	13	0	0	0
	CDH74	4897	0	0	0	17	175	509	1931	1770	473	22	0	0
	CDH80	1969	0	0	0	2	36	147	832	782	169	1	0	0
	WSAvg	9.1	8.1	8.0	9.4	10.7	10.9	9.8	8.6	8.7	8.8	9.3	8.5	7.9
	PrecAvg	15.3	0.6	0.2	0.8	1.4	3.0	2.7	2.4	1.5	1.2	0.7	0.5	0.6
Precipitation	PrecMax	23.2	1.6	0.5	1.2	2.8	5.8	6.3	5.7	4.4	3.6	4.9	1.5	3.4
	PrecMin	10.6	0.2	0.0	0.3	0.4	1.3	0.9	0.7	0.0	0.1	0.1	0.0	0.0
	PrecStd	3.4	0.4	0.2	0.2	0.7	1.3	1.5	1.4	1.1	0.8	1.0	0.4	0.7
	PrecStd	3.4	0.4	0.2	0.2	0.7	1.3	1.5	1.4	1.1	0.8	1.0	0.4	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	50.2	64.4	76.2	84.2	89.8	97.5	96.9	90.6	77.3	62.1	46.4
		MCWB	39.4	40.9	48.4	52.5	57.5	62.2	63.7	62.7	60.0	54.6	47.7	37.8
	2%	DB	39.9	42.2	56.6	69.4	79.0	84.2	92.8	92.4	85.3	68.8	53.9	40.3
		MCWB	35.3	35.7	43.7	50.6	56.7	61.4	63.5	61.7	58.2	51.7	43.2	34.8
	5%	DB	36.7	37.8	50.7	63.9	74.3	79.8	88.9	88.4	79.7	62.6	47.8	36.5
		MCWB	33.6	33.9	40.7	47.3	54.5	60.0	63.1	60.8	56.8	48.3	39.3	32.7
	10%	DB	32.9	33.9	45.0	58.8	69.1	75.7	84.8	84.3	73.7	57.5	42.7	32.9
		MCWB	30.5	31.4	37.5	45.0	52.2	58.4	62.5	60.1	54.6	45.9	36.5	30.2
	0.4%	WB	39.4	41.3	48.6	54.7	61.0	68.4	70.7	68.1	63.6	56.5	48.4	38.6
		MCDB	44.4	49.3	62.3	72.4	77.2	81.8	84.4	84.4	81.8	73.1	60.3	45.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	36.6	44.7	51.2	57.9	64.3	67.8	65.2	60.3	52.5	44.0	35.3
		MCDB	39.2	41.3	55.3	66.1	74.5	77.3	82.5	81.5	77.8	66.2	52.9	39.9
	5%	WB	33.6	33.9	41.1	48.7	55.8	61.9	65.8	63.5	58.1	49.3	40.1	32.8
		MCDB	36.5	37.4	49.5	62.0	70.5	74.7	81.9	80.4	75.2	60.5	46.6	36.2
	10%	WB	30.5	31.4	37.9	45.9	53.7	60.0	64.0	61.9	55.9	46.8	36.9	30.2
		MCDB	33.0	34.1	44.5	57.3	66.3	72.3	79.7	79.1	71.9	56.2	42.2	33.1
Mean Daily Temperature Range	5% DB	MDBR	21.7	22.1	21.9	25.1	26.4	25.3	29.8	30.7	29.2	25.6	22.3	21.6
		MCDBR	22.6	24.1	30.1	35.7	35.8	33.8	37.0	38.8	40.8	36.0	29.5	24.0
		MCWBR	18.6	18.6	18.3	19.2	17.2	15.3	14.2	14.8	18.5	20.0	19.4	19.2
	5% WB	MCDBR	22.3	23.3	28.4	32.3	30.3	28.2	30.5	31.0	35.0	32.8	28.5	23.5
		MCWBR	18.6	18.5	18.0	17.9	15.2	14.5	13.7	12.9	16.6	19.0	19.3	19.1
		MCWBR	18.6	18.5	18.0	17.9	15.2	14.5	13.7	12.9	16.6	19.0	19.3	19.1
Clear-Sky Solar Irradiance	taub		0.247	0.252	0.275	0.327	0.348	0.351	0.359	0.373	0.326	0.285	0.257	0.239
	taud		2.462	2.520	2.533	2.386	2.372	2.420	2.409	2.340	2.440	2.545	2.523	2.477
	Ebn at Noon		269	295	303	292	289	288	284	274	278	276	263	257
	Edh at Noon		19	23	27	35	37	35	35	36	29	22	18	16
All-Sky Solar Radiation	RadAvg		422	741	1145	1559	1865	2022	2187	1795	1299	820	474	332
	RadStd		36	45	62	86	133	126	123	126	109	81	47	23

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page